

Coordinate Geometry

Worksheet 1 (Non-Calculator)

1. W14/qp12/q25

P is $(-4, 4)$ and Q is $(3, -2)$. M is the midpoint of PQ .

- a. Find the coordinates of M . [1]
- b. Find the gradient of the line PQ . [1]
- c. Q is the midpoint of the line PQR .
 - i. Find the coordinates of R . [2]
 - ii. Write down the value of $\frac{PM}{MR}$. [1]

2. S17/qp12/q16

A is the point $(0, 3)$, B is the point $(1, 5)$ and C is the point $(p, -1)$.

- a. Find the equation of the line AB . [2]
- b. The gradient of the line BC is $-\frac{3}{4}$. Find the value of p . [2]

3. W17/qp11/q16

The coordinates of P and Q are $(0, 7)$ and $(10, -1)$.

- a. Find the coordinates of the midpoint of PQ . [1]
- b. The length of PQ is N units, where N is an integer. Find N . [2]

4. W17/qp12/q20

The coordinates of P and M are $(-3, 10)$ and $(0, 4)$.

- a. Find the gradient of the line PM . [1]
- b. Find the equation of the line PM . [1]
- c. M is the midpoint of PQ . Find the coordinates of Q . [2]

5. W18/qp12/q20

P is the point $(-3, 4)$, Q is the point $(5, 1)$.

- a. M is the midpoint of PQ . Find the coordinates of M . [1]
- b. Find the gradient of PQ . [1]
- c. R is the point $(-6, 0)$, O is the point $(0, 0)$. Which of the points, R or P , is closer to O ?
Show your working. [2]

6. W19/qp12/q18

The vertices of a triangle are $A(7, 0)$, $B(-1, 6)$ and $C(-1, -4)$.

- a. Show that $AB = BC$. [3]
- b. Find the area of triangle ABC . [2]

7. W19/qp23/q7 (0580)

Find the gradient of the line that is perpendicular to the line $2y = 3 + 5x$. [2]

8. W19/qp23/q12 (0580)

A is the point $(2, 1)$ and B is the point $(9, 4)$. Find the length of AB . [3]

9. W19/qp23/q13 (0580)

A straight line joins the points $(3k, 6)$ and $(k, -5)$. The line has a gradient of 2.

Find the value of k . [3]

10. W21/qp23/q18 (0580)

Find the equation of the straight line that passes through the points $(2, -2)$ and $(3, 10)$.

Give your answer in the form $y = mx + c$. [3]

Answer Key:

1. (a) $(-1/2, 1)$ (b) $6/7$ (c)(i) $(10, -8)$ (ii) $1/3$
2. (a) $y = 2x + 3$ (b) 9
3. (a) $(5, 3)$ (b) 164
4. (a) -2 (b) $y = -2x + 4$ (c) $(3, -2)$
5. (a) $(1, -1/2)$ (b) $-3/8$ (c) P, OP=5, OR=6
6. (a) AB=BC=10 (b) 40
7. -0.4
8. 7.62
9. 2.75
10. $y = 12x - 26$